V2H tool for self-sufficiency analysis: selected residential use cases

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Abstract. *This paper quantifies the contribution of Vehicle-to-Home (V2H) technology to residential energy self-sufficiency, through simulation. The dispatch that a Home Energy Management System (HEMS), the control layer of the household, would apply to a photovoltaic (PV) installation, an electric vehicle (EV) and a stationary Battery Energy Storage System (BESS) is computed by a Mixed-Integer Linear Programme (MILP) maximising the Self-Sufficiency Rate (SSR), i.e. the share of household demand covered by local generation and storage. Eight controlled use cases (UCs) are analysed, ranging from a grid-only baseline to the full PV+BESS+V2H system. Results demonstrate that an EV with unidirectional smart charging contributes zero SSR improvement in every UC, while V2H raises the SSR by 14.3 percentage points in a PV-only system and by 3.2 percentage points in a system with a BESS also included. The BESS alone outperforms the V2H-capable EV by 33.0 percentage points because, unlike the vehicle, it is never reserved for mobility. Finally, the UCs are examined under Slovenia's new energy-sharing mechanism (slov. souporaba): the UC that maximises individual self-sufficiency exports no surplus and thus has nothing to share, meaning that individual optimisation reduces the energy available for community sharing. This trade-off is quantified, motivating future community-scale optimisation.*

Key words: *Vehicle-to-Home, MILP, HEMS, BESS, self-sufficiency, energy sharing, souporaba*

1 Introduction

The rapid deployment of residential photovoltaic systems has transformed European households from passive consumers into active prosumers. According to the International Energy Agency, global electric car sales exceeded 20 million in 2025, one in four new cars sold, with Europe recording the strongest growth among the major markets [1]. This convergence of rooftop PV and EV adoption raises a central question: can the electric vehicle serve as flexible household energy storage, not just as a transport device?

Vehicle-to-Home (V2H) technology enables bidirectional energy exchange between an EV and the home, turning the vehicle battery into a temporal storage asset that can, for example, absorb solar surplus at midday and return it during the evening peak, thereby providing flexibility to the household. Throughout this paper, V2H denotes this bidirectional operation, which includes smart (unidirectional) charging as a subset. This capability has attracted growing research interest; however, a systematic and experimentally controlled analysis of V2H's marginal contribution to self-sufficiency, isolated from the effect of unidirectional charging alone, remains rare in the literature [3].

Many existing studies either compare a full V2H system against a grid-only baseline, mixing the EV charge load and the V2H discharge benefit into a single number, or use cost minimisation as the primary optimisation objective, which does not directly maximise self-sufficiency [4]. A prior heuristic simulation tool from the same research group, V2Hsim, demonstrated V2H strategies across selected use cases [2]; the present work extends it by formulating an optimal Mixed-Integer Linear Programme (MILP) and isolating the contribution of bidirectionality through a controlled matrix of eight use cases (UCs), presented in Section 2.2.

This paper provides three contributions. First, a MILP formulation that maximises the self-sufficiency rate (SSR) directly, through a linearised auxiliary-variable technique which is shown to outperform indirect proxy objectives (Section 2.3). Second, an experimental matrix of eight UCs that activates the system components one by one, from a grid-only baseline to the full PV+BESS+V2H system; because unidirectional and bidirectional EV operation are separated in two distinct system contexts (PV-only and PV+BESS), the marginal contribution of each component can be quantified individually. Third, an analysis of the results under Slovenia's energy-sharing regulation (slov. souporaba, ZOEE-A, July 2026), showing that the individually optimal dispatch eliminates the exported surplus on which energy sharing relies [6]. This work is conducted within the EV4EU Horizon Europe project [5].

A distinction runs through the whole paper: self-consumption measures how much of the local PV production is used on site, while self-sufficiency measures how much of the household demand is covered locally. The two are often blurred in commercial and even academic literature. An EV that absorbs the midday surplus greatly increases self-consumption, yet if that energy leaves with the vehicle it covers no household demand and self-sufficiency is unchanged. The distinction matters for policy: subsidy schemes or building codes that credit any EV charging infrastructure toward “energy independence” targets may reward unidirectional equipment that delivers no measurable self-sufficiency benefit, while V2H-capable equipment receives no differentiated treatment.

2 System and method

2.1 System architecture

This study is simulation-based: no physical HEMS is implemented. We model the dispatch that a Home Energy Management System (HEMS), the control layer of the house, would apply to four components connected at a single domestic node: a 6 kWp photovoltaic generator, a 60 kWh EV connected through a bidirectional charging station, a 10 kWh stationary BESS, and the grid connection. The 6 kWp PV size sits within the realistic range for residential European installations without a heat pump, typically 6 to 10 kWp [8]; the lower end of the range is chosen deliberately so that an oversized PV system does not mask the BESS and V2H contributions. The household load follows the Standard Family archetype [7] (21.7 kWh/day) with characteristic peaks at 07:00 (1.5 kW, before solar ramp-up) and 19:00 (2.5 kW, after solar shutdown). This temporal mismatch between PV production (concentrated 09:00–16:00) and residential demand (mornings and evenings) defines the storage coordination problem.

Tariffs follow GEN-I's Aktivni cenik elektrike za dom, the active time-of-use tariff for Slovenian residential customers in effect since 6 February 2026 [9]: 0.0426 EUR/kWh during the low-season solar window (09:00–16:00, March to October), 0.1438 EUR/kWh during standard hours, and 0.2195 EUR/kWh during the evening peak (18:00–21:00), all VAT-inclusive. A feed-in rate of 0.05 EUR/kWh is assumed for grid export, consistent with typical Slovenian net-billing arrangements. PV profiles are derived from PVGIS SARAH3 for Ljubljana in June (34.9 kWh/day at 6 kWp). Figure 1 presents the methodology pipeline (a) and the UC matrix (b); the model outputs comprise the SSR, the net daily cost, the exported energy and the dispatched V2H energy, the latter quantifying the flexibility the vehicle delivers to the house.

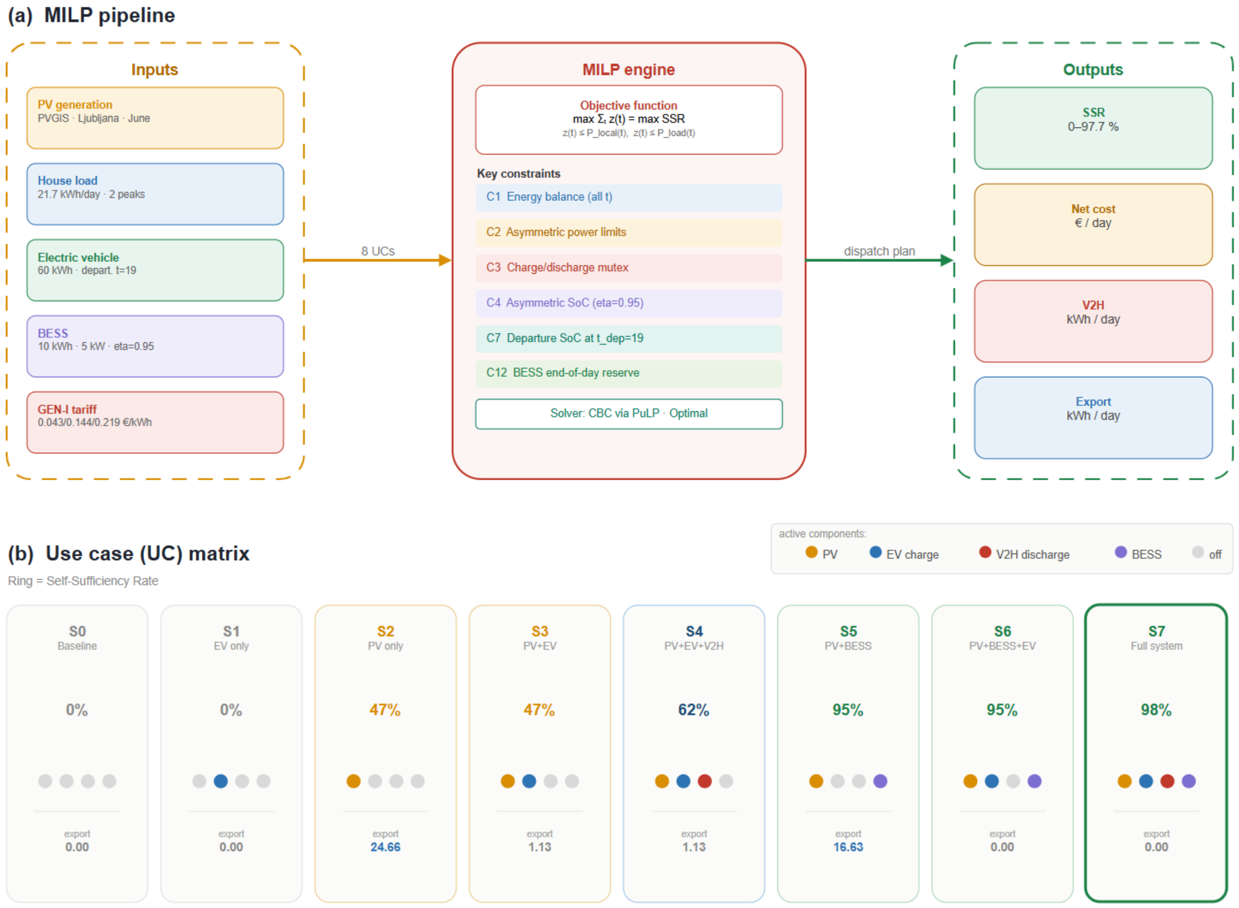

Fig. 1. (a) MILP pipeline: inputs, optimisation engine and outputs. (b) UC matrix: SSR shown as a progress ring, colored dots denote active components, daily export (kWh) below each card.

2.2 Use cases investigated

Eight UCs, labelled S0 to S7, are constructed by activating the system components one at a time (Table 1). Beyond the six UCs of a standard component comparison (baseline, PV, EV, V2H, BESS, full system), two additional UCs make unidirectional versus bidirectional EV operation an explicit experimental variable in two system contexts: S3 versus S4 isolates the contribution of V2H in a PV-only system, and S6 versus S7 isolates it again when a BESS is present. All eight UCs share identical input data — the same PV profile, load profile, EV parameters and tariff structure — so any difference in SSR or cost is a structural consequence of component activation, not of input variation.

In all EV-active UCs the vehicle follows the same availability window: it is plugged in at home from midnight until its departure at 19:00 and is away for the rest of the evening, its state of charge being held constant while absent.

Table 1. The eight controlled use cases.

UC	Active components
S0	Baseline (grid only)
S1	EV, smart charging only
S2	PV only
S3	PV + EV smart charging
S4	PV + EV + V2H
S5	PV + BESS
S6	PV + BESS + EV smart charging
S7	Full system: PV + BESS + EV + V2H

2.3 MILP formulation and SSR objective

The SSR is the proportion of the household's daily energy demand met by local sources (PV generation plus storage discharge) without drawing from the grid: an SSR of 100%

means the house never imports electricity over the 24-hour horizon. At every hourly slot t, only the local energy actually consumed by the house counts; local energy beyond the instantaneous demand is stored or exported and does not contribute:

$$\mathrm{SSR} = \frac{\sum_{t=1}^{24} \min(P_{\mathrm{loc}}(t), P_{\mathrm{load}}(t))}{\sum_{t=1}^{24} P_{\mathrm{load}}(t)} \qquad (1)$$

where $P_{loc}(t)$ is the locally available power (PV plus BESS and EV discharge) and $P_{load}(t)$ the household demand. The min() function is non-linear, while a MILP requires linear expressions. The standard linearisation introduces an auxiliary variable z(t) for each slot, bounded by both quantities:

$$z(t) \leq P_{\mathrm{loc}}(t), \quad z(t) \leq P_{\mathrm{load}}(t) \quad \forall t \qquad (2)$$

The optimiser maximises Σ z(t); since z(t) can exceed neither bound, it is pushed to the smaller of the two at every slot, recovering (1) in linear form. Two secondary terms complete the objective: a soft penalty (10 EUR per kWh of shortfall) discouraging the BESS from ending the day depleted, and a small economic tiebreaker (weight 0.001) applied only between plans of identical SSR. An earlier version used a weighted proxy objective instead, rewarding V2H and BESS discharge while penalising grid import; it underperformed the direct SSR objective by 5.1 percentage points in UC S3, because rewarding storage for discharging is not the same as rewarding it for reducing grid dependency when the house needs it.

2.4 Constraint design

The model enforces twelve constraint families; the most relevant ones, listed in Fig. 1(a), are formalised below for each hourly slot t. C1 balances power at the domestic node:

$$P_{\mathrm{PV}} + P_{\mathrm{imp}} + P_{\mathrm{dis}} = P_{\mathrm{load}} + P_{\mathrm{ch}} + P_{\mathrm{exp}} \quad \forall t \qquad (3)$$

where P_{ch} and P_{dis} aggregate the EV and BESS charging and discharging power and P_{imp}, P_{exp} are grid import and export. C2 and C3 bound the EV power asymmetrically and forbid simultaneous charging and discharging through a binary variable b(t):

$$0 \leq P_{\mathrm{EV,ch}}(t) \leq 7.4\, b(t)$$

$$0 \leq P_{\mathrm{EV,dis}}(t) \leq 11\,(1 - b(t)) \qquad (4)$$

The two directions have different limits because they use different hardware paths: charging is limited to 7.4 kW by the 32 A single-phase AC wall box, while V2H discharge flows through the bidirectional charging station rated at 11 kW. V2H thus requires an EV and a charging station that both support bidirectional operation; in the results below the discharge limit is never binding. C4 applies the efficiency η = 0.95 once per direction,

$$\mathrm{SoC}(t+1) = \mathrm{SoC}(t) + \eta\, P_{\mathrm{ch}}(t)\Delta t - P_{\mathrm{dis}}(t)\Delta t/\eta \quad (5)$$

so that a full charge–discharge cycle loses energy twice; applying η only once per cycle, as an earlier version did, underestimates round-trip losses by about 10% and makes V2H discharge appear artificially cheap. C7 enforces mobility: as a working assumption, the EV battery must hold at least 80% state of charge at the departure slot:

$$\mathrm{SoC}_{\mathrm{EV}}(t_{\mathrm{dep}}) \geq 0.8\, E_{\mathrm{EV}}, \quad t_{\mathrm{dep}} = 19 \qquad (6)$$

Anchoring the requirement at the actual departure slot matters: an earlier version checked the 80% target at midnight instead, allowing plans in which the vehicle left at 19:00 nearly empty and "recharged" while absent; anchoring at t_{dep} guarantees that every proposed plan is physically executable. C12 is the BESS end-of-day soft reserve already introduced in Section 2.3:

$$\mathrm{SoC}_{\mathrm{B}}(24) \geq \mathrm{SoC}_{\mathrm{B}}(0) - \delta, \quad \delta \geq 0 \qquad (7)$$

with δ penalised in the objective. The remaining families protect battery health with a 30% state-of-charge floor for the EV and the BESS, hold the EV state of charge constant while the vehicle is away, cap storage discharge at the instantaneous house demand, and block grid export while local storage discharges to serve the house. The model is solved with CBC, an open-source MILP solver, through the PuLP library in Python.

3 Results

All eight UCs return Optimal solver status, and the BESS end-of-day reserve (7) is never violated (δ = 0 in all BESS-active UCs), confirming adequate dimensioning for June conditions. Figure 2 shows the SSR of all UCs. UC S7, the full system, reaches the highest SSR (97.7%); Figure 3 presents its full 24-hour dispatch, which the optimiser finds without any fixed priority rule. This is the central result of this study.

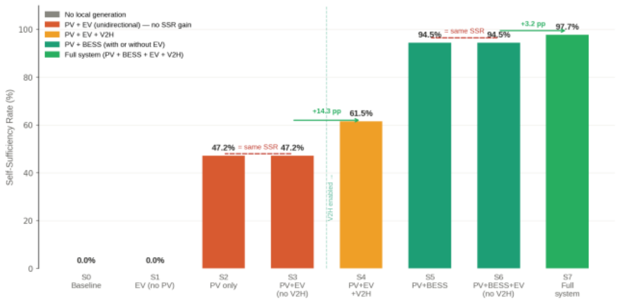

Fig. 2. SSR for all eight UCs. Dashed brackets mark identical SSR pairs (S2 = S3; S5 = S6); green arrows quantify the V2H marginal gains.

3.1 Unidirectional charging: zero SSR contribution

The key result is S3 = S2 = 47.2% and S6 = S5 = 94.5%: in both pairs, adding the EV in charging-only mode leaves the SSR unchanged despite substantially reducing grid export, from 24.66 kWh to 1.13 kWh between S2 and S3, and from 16.63 kWh to zero between S5 and S6. The EV absorbs the PV surplus effectively but cannot return it to the house, and Eq. (1) gives no credit to charging, which draws power from the node rather than supplying it. Self-consumption therefore rises strongly while self-sufficiency does not move: the two metrics are fundamentally different, and equipment optimised for one may have no effect on the other. In practical terms, a smart charger that eliminates export may appear to improve the energy performance of the house while delivering zero self-sufficiency benefit.

3.2 V2H marginal contribution

Enabling V2H in a PV-only system (S3 to S4) raises the SSR by 14.3 percentage points, from 47.2% to 61.5%. The vehicle discharges 3.1 kWh during the evening, partially covering the 2.5 kW peak that S3 imported entirely from the grid. Since no BESS is present in S4, the EV battery is, within this UC, the only device able to shift energy in time, absorbing solar surplus at noon and returning part of it in the evening. With a BESS already covering most of the evening demand (S6 to S7), V2H adds a more modest 3.2 percentage points, deploying 1.8 kWh at exactly the slot where the BESS cannot simultaneously cover the full load and preserve its end-of-day reserve. In both configurations the 80% departure requirement (6) is met exactly at 19:00: the mobility constraint is always binding, and the optimiser never sacrifices mobility for self-sufficiency.

3.3 BESS dominance

UC S5 (PV+BESS, no EV) achieves 94.5% SSR, outperforming S4 (PV+EV+V2H, no BESS) by 33.0 percentage points, although the EV battery is six times larger than the BESS. Two structural reasons explain this. The first is availability: the departure constraint (6) reserves 48 kWh of the EV battery for mobility, leaving at most 18 kWh above the 30% floor for V2H, whereas the BESS cycles its full 8 kWh usable range unconditionally every day. The second is timing: the vehicle leaves at 19:00, precisely when the evening peak begins, so the hours in which the house most needs local energy are the hours in which the EV is least available. A smaller battery that is always present therefore outperforms a much larger battery that is partially committed to mobility.

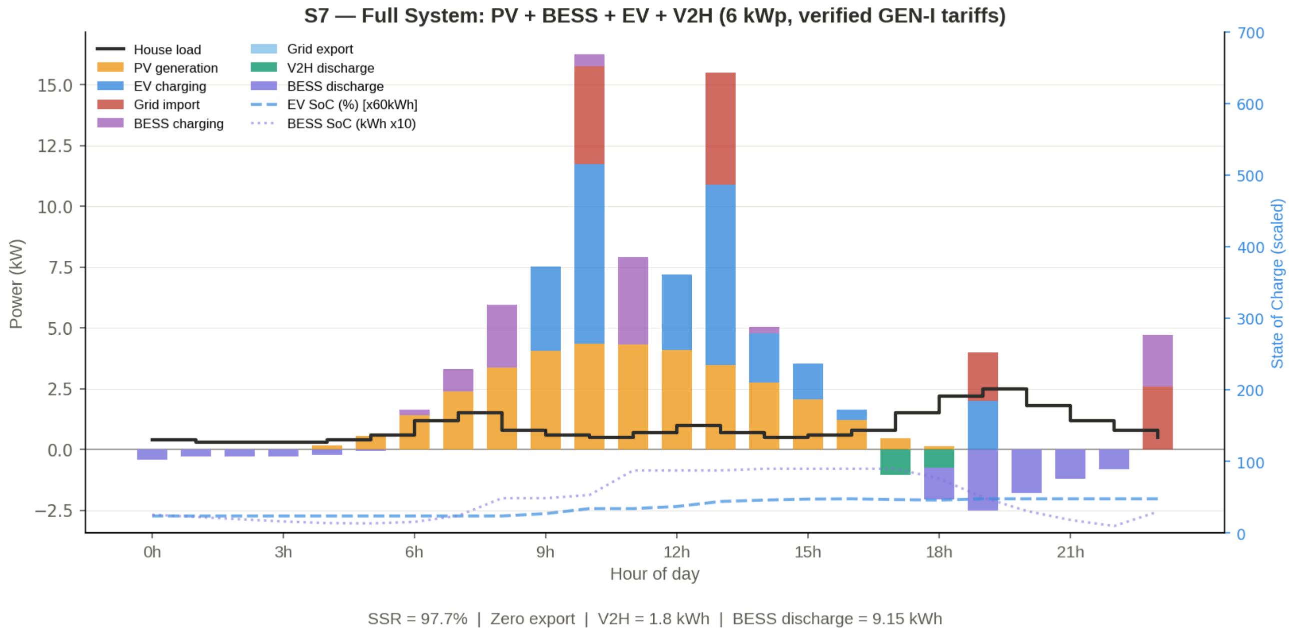

Fig. 3. 24 h power flows and SoC curves, UC S7 (full system). The MILP coordinates three assets without fixed priority rules: the BESS covers the night load, the EV charges from midday solar, V2H intervenes during the evening peak (1.8 kWh). Zero export; SSR = 97.7%.

4 Discussion

4.1 Investment hierarchy for June conditions

For the June inputs of Section 2.1, the results establish a clear investment priority sequence. Photovoltaic generation is the essential first investment: the SSR rises from 0% to 47.2% and the net daily cost falls from 3.15 to 0.89 EUR under the GEN-I tariff, and no subsequent investment delivers value without solar surplus to work with. The BESS is the dominant self-sufficiency investment: the SSR rises from 47.2% to 94.5% and the net cost turns negative (−0.18 EUR/day) as the household becomes a net exporter, the BESS completing one full cycle per day on this load profile. V2H delivers the final 3.2 percentage points to 97.7% at a net cost of 1.18 EUR/day; the 1.36 EUR/day increase over S5 pays for approximately 25 kWh of daily mobility, i.e. roughly 0.05 EUR per kWh of driving energy, well below petrol-equivalent cost.

4.2 Year-round overview

The component ranking above is confirmed outside June: re-running UCs S2, S5 and S7 for representative March and December days preserves the ordering, while the underlying numbers shift considerably. PV-only self-sufficiency falls sharply with the season, from 47.2% in June to 33.1% in March and 19.4% in December, since a PV-only system has no way to bridge a shortage of daylight: December irradiance in Ljubljana is roughly 85% lower than in June (5.3 versus 34.9 kWh/day at 6 kWp). The full system is far more resilient: its SSR only falls from 97.7% to 96.3% in both March and December, because the MILP remains free to charge the BESS from the grid in the cheapest tariff window. What degrades is cost, not self-sufficiency: the net daily cost of S7 rises from 1.18 EUR in June to 2.04 EUR in March and 3.08 EUR in December, as more of the charging energy is imported instead of coming from free solar.

4.3 Souporaba: individual optimisation versus community sharing

The souporaba mechanism (ZOEE-A, in force July 2026) enables virtual bilateral energy sharing between a transmitter prosumer and a receiver household through Slovenia's nationwide smart-meter network, independently of geographic proximity. Only the energy commodity cost is transferred; network fees remain unchanged for the receiver. Settlement is at 15-minute resolution, with one important consequence: the transmitter's surplus benefits the receiver only if the receiver consumes in the same 15-minute interval, so a midday PV surplus is only valuable to receivers with midday consumption.

Re-examining the UCs from the perspective of a receiver reveals a conflict. S7, the individually optimal UC, exports nothing and thus provides nothing to share; S5 exports 16.63 kWh per day, the best individual–community balance in the matrix. As a household progresses from S5 to S7, the surplus that souporaba is designed to redistribute disappears: a receiver who signed a sharing contract with a neighbour operating S5 would see the allocation drop to zero the day the neighbour installs a smart-charging EV. Under the current framework, individual optimisation and community sharing are competing objectives, a structural tension also identified in broader community-energy research [6].

The mechanism does provide benefits, but under identifiable conditions on the transmitter side: (i) PV generation exceeding what the local load and storage can absorb, as in S2 and S5; (ii) the surplus season, since the shareable energy concentrates in the high-irradiance months; and (iii) dispatch that is not individually SSR-optimal, by lack of equipment or by deliberate choice. Souporaba thus redistributes the surplus of imperfectly optimised or generously dimensioned prosumers, and its potential shrinks precisely as HEMS adoption grows.

This motivates a community-scale MILP coordinating dispatch across multiple households, with a community sharing term alongside individual SSR. The trade-off is already quantifiable from our results: moving deliberately from S7 to S5 costs the household 3.2 percentage points of SSR while freeing 16.63 kWh of daily export for sharing. Regulatory incentives compensating transmitters for maintaining shareable surplus would be needed to make this trade-off rational [6].

4.4 Limitations and scope

Three limitations apply. First, the MILP assumes perfect 24-hour forecasts of PV production and load; the SSR values reported are thus an upper bound on what a forecast-dependent controller (rolling-horizon or robust) could achieve. Second, the 24-hour horizon resets each day with no inter-day coupling of the BESS state of charge, appropriate for comparing UCs but requiring extension for continuous operation. Third, the single-household scope, necessary to isolate the V2H contribution, does not capture network-level effects such as transformer loading or voltage constraints if many households on the same low-voltage feeder adopted similar patterns simultaneously.

5 Conclusion

Three results are established. First, an EV with unidirectional smart charging contributes zero SSR improvement in every configuration tested; the result is structural, since charging never supplies the house, and it shows that self-consumption and self-sufficiency are distinct objectives. Second, under the conditions studied (Ljubljana PV profile, Standard Family load, 60 kWh EV departing at 19:00 with an 80% SoC requirement, 10 kWh BESS), V2H adds 14.3 percentage points of SSR in a PV-only system and 3.2 percentage points in a full system, while the BESS is the dominant investment at +47.3 percentage points over PV alone; the ranking is preserved in March and December. Third, the SSR-optimal household exports nothing and thus has nothing to share under souporaba: under the current regulation, individual optimisation and community sharing are competing objectives, and shareable surplus exists only where dispatch or dimensioning departs from the individual optimum. The MILP formulation and the controlled UC methodology developed here provide the computational foundation for community-scale optimisation jointly considering the self-sufficiency of multiple households, their shared surplus, and their participation in flexibility services.

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